



Shapes of the saturnian icy satellites and their significance

P.C. Thomas^{a,*}, J.A. Burns^a, P. Helfenstein^a, S. Squyres^a, J. Veverka^a, C. Porco^b, E.P. Turtle^c,
A. McEwen^d, T. Denk^e, B. Giese^f, T. Roatsch^f, T.V. Johnson^g, R.A. Jacobson^g

^a Center for Radiophysics and Space Research, Cornell University, Ithaca, NY 14853, USA

^b Space Science Institute, 4750 Walnut Street, Boulder, CO 80301, USA

^c Johns Hopkins University Applied Physics Laboratory, 11100 Johns Hopkins Rd., Laurel, MD 20723, USA

^d Department of Planetary Sciences, University of Arizona, Tucson, AZ 85721, USA

^e Institut für Geologische Wissenschaften, Freie Universität, 12249 Berlin, Germany

^f Institute of Planetary Research, German Aerospace Center, Rutherfordstrasse 2, 12489 Berlin, Germany

^g Jet Propulsion Laboratory, California Institute of Technology, 4800 Oak Grove Drive, Pasadena, CA 91109, USA

Received 21 August 2006; revised 9 March 2007

Available online 4 April 2007

Abstract

The sizes and shapes of six icy saturnian satellites have been measured from Cassini Imaging Science Subsystem (ISS) data, employing limb coordinates and stereogrammetric control points. Mimas, Enceladus, Tethys, Dione and Rhea are well described by triaxial ellipsoids; Iapetus is best represented by an oblate spheroid. All satellites appear to have approached relaxed, equilibrium shapes at some point in their evolution, but all support at least 300 m of global-wavelength topography. The shape of Enceladus is most consistent with a homogeneous interior. If Enceladus is differentiated, its shape and apparent relaxation require either lateral inhomogeneities in an icy mantle and/or an irregularly shaped core. Iapetus supports a fossil bulge of over 30 km, and provides a benchmark for impact modification of shapes after global relaxation. Satellites such as Mimas that have smoother limbs than Iapetus, and are expected to have higher impact rates, must have relaxed after the shape of Iapetus was frozen.

© 2007 Elsevier Inc. All rights reserved.

Keywords: Saturn, satellites; Satellites, shapes; Interiors

1. Introduction

Saturn's system of satellites includes seven approximately ellipsoidal objects and dozens of smaller ones with irregular shapes. The larger satellites display a wide range of surface features, albedos, and mean densities. Ellipsoidal objects may be equilibrium shapes determined by their mean density, mass distribution, and spin rates (Chandrasekhar, 1969; Dermott, 1979), and as such their shapes may contain information on their interiors. Likewise, departures from equilibrium forms may reveal information on processes that complicate or defeat tendencies to relax to equilibrium shapes. Cassini data permit measurement of the shapes and masses of most of the

ellipsoidal satellites sufficiently well to test models of the interior structure and the extent of relaxation.

In this work we first review the method of shape determination and evaluation. Next we report the measurement results for six satellites (Fig. 1). We then examine how close each moon is to an equilibrium shape. Then we view the results in toto, and finally evaluate the significance of departures from equilibrium forms. We do not consider the largest satellite Titan, as ISS cannot detect the satellite's surface limb and shape measurement requires a global network of control points, which are not yet available.

2. Methods and data

2.1. Shape measurement

The overall shapes are found by measuring limb positions in the ISS images. The process involves these steps: (1) Mea-

* Corresponding author.

E-mail address: pct2@cornell.edu (P.C. Thomas).

of Enceladus. Those with irregularly-shaped cores, and perhaps some variation in ice shell densities might be consistent with the shape if the core mean density is less than $\sim 2200 \text{ kg m}^{-3}$. While equilibration closer to Saturn might explain the shape better, it would still need to support this topography in current conditions. The tectonic patterns suggest polar flattening of an indeterminate amount. If Enceladus were differentiated and relaxed, and then flattened to the present form, it would require effects at depth of nearly hemispherical extent; the amplitude of topography of the south polar tectonic province is only a small fraction of the differences between homogeneous and highly differentiated shapes.

In sum, the shape of Enceladus suggests that if it is differentiated it has a large, low-density core, the core itself might have inhomogeneities in density or shape, and an outer ice shell may be inhomogeneous in density by 10% or more on nearly hemispherical scales. Fully relaxed, radially differentiated, laterally homogeneous models do not explain the shape.

3.4. Tethys

Tethys' shape is consistent with that of a relaxed object in hydrostatic equilibrium, although the satellite's roughness (Giese et al., 2006) and the less-than-desirable image-coverage to date create large uncertainties in the results. The supported topography for a homogeneous model is 0.77, +0.74, –0.22 km. Tethys' shape is thus almost consistent with the interior being homogeneous. The equilibrium shape of any differentiated model differs little from the homogeneous case because the very low mean density requires that any core that is present must be very small. For our standard mantle and core densities of 930 and 3000 kg m^{-3} , Tethys has supported topography of 0.55, +0.79, –0.06 km. The core would constitute 0.06 of the satellite's mass and have a radius of 145 km.

3.5. Dione

Dione's shape is consistent with that of a relaxed object, and the $(a - c)$ is marginally consistent with the homogeneous case. The homogeneous model supports 0.54, +0.49, –0.44 km of topography. Our standard core model would support 0.11 km, +0.6 km, and is thus our best differentiated model. (For the standard core model, all shape uncertainties increase the supported topography from the nominal value, thus the single error value.) This model has a core mass ratio of 0.54, and a core radius of 360 km.

3.6. Rhea

Rhea's $(a - c)$ is consistent with the satellite being in hydrostatic equilibrium, but the placement of limb profiles results in significant uncertainties in the a axis and in the relative values of the b and c axes; the nominal solution has $c > b$, which would imply lateral inhomogeneities. A homogeneous Rhea would support 1.02, +1.74, –0.16 km of topography. Our standard mantle/core combination has a core mass ratio of 0.36, and supports 0.94, +1.12, –0.71 km of topography.

While the measurements may permit a shape consistent with a relaxed, differentiated structure, the uncertainties also allow for substantially different interior models. For these examples, the uncertainty in $(a - c)$ drives the highest supported topography.

3.7. Iapetus

For decades—even centuries, the major Iapetus puzzle was how the satellite's leading and trailing sides acquired albedos that differed by a factor of ~ 10 . Four years before Cassini's arrival, Denk et al. (2000), using Voyager images, found a new conundrum: Iapetus has a decidedly non-spherical shape, something not expected given the apparently relaxed nature of objects that are considerably smaller than this large outer moon. Images obtained in the first year of Cassini's orbital tour have documented this non-spherical shape and have revealed significant global tectonism on Iapetus.

Iapetus is markedly non-spherical; the best-fit triaxial shape is 748.8, 743.2, 712.4 km, with the long axis at 320° W longitude. An oblate spheroid of axes 747.4 by 712.4 km is within the allowed residuals. We treat Iapetus as an oblate spheroid (Table 1). The gap in limbs in the 200° – 280° W longitude range, and the low resolution of most images between 280° and 360° W raises some concern about the fit's accuracy, but subdivisions of the data yield shapes within a few km of our adopted values, and errors would have to be 20 km or more to qualitatively alter the conclusions. The satellite's density and spin rate predict an axial difference of only ~ 10 m, not the measured difference of 35 km. Because of the absolute value of Iapetus' limbs roughness (Fig. 3a), we might suspect that Iapetus has never relaxed. However, the relative roughness, less than 1% of the mean radius, Fig. 3b (also see Fig. 1), indicates that Iapetus' shape indeed is an oblate spheroid, probably a result of relaxation, or approach to relaxation. Such relaxation and subsequent freezing of the shape would have to have occurred at a higher spin rate.

Iapetus has topography that dwarfs that on Enceladus (Fig. 7). Two key departures from the reference spheroid on Iapetus are the equatorial ridge (Porco et al., 2005; Denk et al., 2005) and large craters (Fig. 7). The equatorial ridge is crossed by limb profiles at different locations and at different angles (see Fig. 2). Removing the equatorial ridge portions from the data reduces the equatorial fit axes by 1.2 km.

Iapetus currently rotates synchronously with its orbital period of 79 days. Because of the satellite's distance from Saturn, despinning in the age of the Solar System has posed some theoretical challenges, requiring a very dissipative interior, with a dissipation factor, Q , < 50 (Peale, 1977). Today's measured shape adds an interesting aspect to this puzzle since the oblate form is consistent with relaxation at a spin period of 16.4 ± 0.9 , -0.8 h for a homogeneous body. (If Iapetus were differentiated with our standard mantle/core densities, the shape would be consistent with a spin period of 15.2 ± 0.8 , -0.7 h.) Furthermore, according to this model, the satellite must have achieved rigidity while at that spin period and have deformed little subsequently. Iapetus, if homogeneous, currently supports 14.3 km of topography; our standard mantle/core model would